# Project Proposal

Team Name: Healthcare - Understanding Chronic Diseases

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# Problem and Goal

Research question:

- Which state-level demographic, socioeconomic and healthcare factors are associated with higher chronic disease rates?

- The importance of this question focuses on understanding how different state-wide resources and demographics can impact chronic disease rates. We use state data to identify patterns and tie it in with chronic disease rates across the United States.

# Data Plan

- Our base dataset is  one that describes chronic disease indicators by US state (taken from data.gov)
    - Supplementary datasets will be acquired to describe aspects of every state from the period 2019-2023 that describe per capita income, healthcare access, insurance coverage, etc. These will act as features to find correlations for disease indicators
- Most data is formatted as a CSV, so the acquisition of data will be simple. Supplementary datasets can be taken from data.gov or from US Census data

# Formalization into data mining

- Reduce the dimensionality of the dataset's chronic disease indicators with SVD and identify clusters of similar states using K-means clustering
- Use Spark or Databricks to integrate geospatial data and features from other datasets into the base dataset
- Use linear regression to identify which features most significantly impact chronic disease indicators across all states
- Use PowerBI to create an interactive dashboard that visualizes disease indicators and their factors overlayed on a map of the US

# Teamwork agreement

Link to the teamwork agreement:
Teamwork Agreement

Software development teamwork model:
Agile/Scrum Model

Git repository:
Repository

Jira:
Project Managment Tool

# Schedule

## Milestone One: Midterm Check
Due: November 11th, 2026
Deliverables: Temporary report containing team progress, issues or roadblocks, and a draft of the work accomplished.

## Milestone Two: Final Report
Due: December 6th, 2026
Deliverables: A final report with data and code containing information about problem context, algorithms, and final dashboard output.

## Milestone Three: Final Presentation
Due: December 8th, 2026
Deliverables: A final presentation to be presented to the class containing problem introduction, experiment results, and final dashboard output.